Ideation Phase
Empathize & Discover

Date	31 January 2026
Team ID	LTVIP2026TMIDS24253
Project Name	**ELECTRICITY CONSUMPTION ANALYSIS**
Maximum Marks	4 Marks

EMPATHY MAP CANVAS

Empathy Map Canvas:
An empathy map is a simple visual tool that captures users' behaviors, needs, and concerns related to electricity consumption. It helps the team understand how energy planners, policy makers, researchers, and general public perceive energy consumption data and regional patterns. In this project, the empathy map is used to understand users who want clear insights into electricity consumption patterns but struggle with complex and scattered energy information. By considering users' goals, challenges, and emotions, the team can design meaningful Tableau dashboards that present energy data clearly, support planning decisions, and enable better resource allocation choices.

EMPATHY MAP TABLES
Table 1: User Segments

User Segment	Primary Role	Key Characteristics	Data Access Level
Energy Planners	Regional energy management	Technical background, decision-making authority	High access
Policy Makers	Government officials	Policy development focus, non-technical	Medium access
Researchers	Academic/industry research	Analytical mindset, detail-oriented	Variable access
General Public	Citizens/consumers	Limited technical knowledge, curiosity	Low access
Students	Educational users	Learning focus, academic requirements	Medium access

Table 2: User Goals & Aspirations

User Type	Primary Goal	Secondary Goals	Success Definition
Energy Planners	Optimize resource allocation	Reduce waste, improve efficiency	Data-driven decisions
Policy Makers	Develop effective energy policies	Public satisfaction, cost reduction	Policy impact evidence
Researchers	Identify consumption patterns	Publish findings, advance knowledge	Pattern discovery
General Public	Understand energy usage	Reduce bills, environmental impact	Personal insights
Students	Learn about energy consumption	Complete assignments, gain skills	Educational achievement

Table 3: User Challenges & Pain Points

Challenge Category	Specific Pain Point	Impact on Users	Frequency
Data Complexity	16,599 records overwhelming	Analysis paralysis	Daily
Geographic Barriers	Regional data scattered	Incomplete understanding	Weekly
Temporal Confusion	Time-based patterns unclear	Poor trend analysis	Monthly
Technical Jargon	Complex terminology	Misinterpretation	Constant
Tool Limitations	Basic visualization tools	Poor insights	Always

Table 4: User Emotions & Feelings

User Situation	Positive Emotions	Negative Emotions	Emotional Triggers
Data Discovery	Curiosity, excitement	Overwhelm, confusion	Large datasets
Pattern Finding	Satisfaction, achievement	Frustration, disappointment	Complex analysis
Decision Making	Confidence, clarity	Uncertainty, anxiety	Important choices
Tool Usage	Empowerment, efficiency	Frustration, helplessness	Poor interfaces
Learning Process	Engagement, growth	Intimidation, boredom	Technical content

Table 5: User Behaviors & Actions

Behavior Category	Current Actions	Desired Actions	Barriers
Data Access	Manual data collection	One-click access	System limitations
Analysis	Spreadsheet manipulation	Interactive visualization	Tool constraints
Decision Making	Gut-feeling decisions	Data-driven choices	Lack of insights
Collaboration	Email sharing	Real-time collaboration	Platform restrictions
Learning	Textbook reading	Interactive exploration	Educational resources

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DETAILED EMPATHY MAP COMPONENTS

Table 6: SAYS (Verbal Expressions)

User Segment	What Users Say	Underlying Meaning	Design Implications
Energy Planners	"I need regional comparisons"	Want geographic insights	Regional visualization
Policy Makers	"Show me policy impact"	Need evidence for decisions	Policy impact analysis
Researchers	"Where are the patterns?"	Pattern	

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Table 2: User Goals & Challenges

User Type	Primary Goal	Main Challenge	Success Definition
Energy Planners	Optimize resource allocation	Complex data analysis	Data-driven decisions
Policy Makers	Develop effective policies	Technical jargon	Policy impact evidence
Researchers	Identify consumption patterns	Scattered data sources	Pattern discovery
General Public	Understand energy usage	Overwhelming information	Personal insights
Students	Learn about energy	Academic complexity	Educational achievement

Table 3: User Emotions & Behaviors

Emotion Category	Positive Feelings	Negative Feelings	Behavioral Impact
Data Discovery	Curiosity, excitement	Overwhelm, confusion	Analysis paralysis
Pattern Finding	Satisfaction, achievement	Frustration, disappointment	Tool abandonment
Decision Making	Confidence, clarity	Uncertainty, anxiety	Poor choices
Tool Usage	Empowerment, efficiency	Frustration, helplessness	Resistance to adoption
Learning Process	Engagement, growth	Intimidation, boredom	Disengagement

Table 4: Design Implications

User Need	Design Solution	Technical Implementation	User Benefit
Simplified Data	Interactive visualizations	Tableau dashboards	Clear insights
Regional Analysis	Geographic comparisons	Map visualizations	Better planning
Pattern Recognition	Trend analysis tools	Time series charts	Pattern discovery
Decision Support	Evidence-based insights	Policy impact analysis	Better decisions
Educational Content	Learning resources	Interactive tutorials	Skill development

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KEY EMPATHY INSIGHTS

Primary User Insights:

- **Users need simplified access to complex energy data**
- **Geographic and temporal patterns are most valuable**
- **Interactive tools significantly improve understanding**
- **Different user segments require different approaches**
- **Emotional barriers impact data adoption**

Design Priorities:

- **Create intuitive, non-technical interfaces**
- **Provide multiple visualization types**
- **Support both detailed and summary views**
- **Enable real-time interaction and exploration**
- **Include educational components for learning**